INCH-POUND

A-A-59780
27 September 2005

COMMERCIAL ITEM DESCRIPTION

PLURAL COMPONENT SPRAY EQUIPMENT FOR PAINT APPLICATION

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) covers plural component paint application equipment. Additional items and or services required under a specific purchasing contract are not identified in this CID. Plural component equipment provides a means of combining the "A" and "B" components of a two-component coating, mixing them at a specified ratio and spraying them through a hand-held spray gun. The unit is typically pneumatically driven.

2. CLASSIFICATION.

2.1 Type 1. Changing the ratio requires that the metering pump displacement pistons and cylinders (lowers) be physically removed and exchanged with a different set of lowers having a different volumetric displacement. The accuracy shall be ± 3%.

2.2 Type 2. Changing the ratio on pumps using pivoting lever arm assemblies requires that the pivot points of the metering pump displacement pistons be fixed to a new location along the lever arm by means of a pin or bolt that passes through a discrete location on the lever arm. The accuracy shall be ± 3%.

2.3 Type 3. Mix ratio electronically selectable in fixed increments to the desired ratio setting. A key shall be required to change the mix ratio. The mix ratio shall be clearly displayed on the proportioner. The accuracy shall be ± 3%.

3. SALIENT CHARACTERISTICS. As a minimum, the plural component unit or "system" shall consist of the following attributes as listed below and discussed in the following paragraphs.

a. Variation of component ratio
b. Metering pump
c. Static mixer
d. Remote mixing station (or block)
e. Heaters for each component
f. 5- to 20-gallon (18.9- to 75.7-liter) feed hoppers or optional 55-gallon (208.2-liter) feed station for each paint component
g. Solvent flush system
h. Metering ratio check station
i. Automatic loss of product shut down for each component

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FSC 4940

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j. Emergency shutoff

k. Overpressure safety shutoff

l. Cycle counter

m. Air control valve reset (for pneumatic systems)

3.1 Variation of component ratio. The ratio of the A and B components shall be set to discrete ratios vice being continuously variable.

3.2 Metering pump. The metering pump shall meter the two components of the coating system at established ratio and deliver it at the required working pressure for spraying.

3.3 Static mixer. The static mixer shall mix the metered coating material for proper curing or hardening reaction. The static mixer shall be easily removable for cleaning or shall be a disposable type.

3.4 Remote mixing station. The remote mixing station includes control and flushing valves with a static mixer. The remote mixing station shall be capable of handling heated but unmixed material over a large transport distance from the metering pump. The remote mixing station must be capable of the following modes:

a. Re circulation of unmixed components and material return to supply bins.

b. Mixing (through static mixer), supplying multiple spray gun lines

c. Isolation of individual (mixed) gun lines.

d. Solvent flushing of individual (mixed) gun lines without impacting operation of spray gun lines still in use.

3.5 Heaters. Heating of material and maintenance of heated conditions shall be accomplished through the use of direct in-line heaters. Heaters shall have variable thermostatic controls. Supply lines may be heated and or insulated. Heaters shall be capable of delivering material at a minimum of 120 °F (48.9 °C) to gun feed lines while unit is operating at maximum rated capacity. In addition, the unit shall be capable of maintaining a minimum material temperature of 77 °F (25 °C) at the spray gun when unit is operating at full capacity at 40 °F (4.4 °C) outside temperature. Heater power requirements shall be dual 220/440 volts at 60 cycles. Heaters shall be capable of safe operation at the maximum system pressure.

3.6 Standard feed system. The standard feed system shall consist of adequate, typically 5- to 20-gallon (18.9- to 75.7-liter), A and B hoppers mounted to the unit and configured for easy detachment for removal and replacement. The hoppers shall be configured with provisions for mechanical agitation equipment when needed. The hopper feed pumps (to main metering pump) shall be capable of supplying material without cavitation while the unit is operating at maximum capacity.

3.7 Optional 55-gallon (208.2-liter) feed stations. The 55-gallon (208.2-liter) feed stations shall consist of the equipment utilized in managing the component feed to the metering pump, including agitation equipment. The drum handling system must be powered to allow operators to manipulate the system without excessive physical labor (for example, pneumatic rams may be used to raise and lower the equipment into the 55-gallon (208.2-liter) drums). If the plural component equipment is mounted in a container, the drum handling and feed system must be fully functional within the container, whether the system is hard mounted within the container, on wheels, or on a skid. The assembly must include two stations, one for each of the unmixed paint components in 55-gallon (208.2-liter) drums. The feed system (to main metering pump) shall be capable of supplying material without cavitation while the unit is operating at maximum capacity.

3.8 Solvent flush system. The solvent flush system shall be capable of flushing the entire plural feed system, to include remote mixing station and individual component feed lines as well as main metering system, without the need for physical line or hose changeovers. The solvent flush system shall be capable of providing pressures between 3000 and 5000 psi (20.7 and 34.5 MPa) depending on hose lengths, number of guns, and material viscosity. Minimum volume of solvent flush system is 1.0 gpm (3.8 lpm).

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3.9 Metering ratio check station. The metering ratio check station shall allow for the verification of proper mixing ratio of the coating system. The ratio check station shall be capable of providing individualized volumes of the unmixed components at the static line or operational pressure. The ratio check station shall have a means to simultaneously control the flow of both unmixed components (e.g., by pneumatically actuated valves or mechanically interlocked valves). The equipment manufacturer shall provide operation documentation that outlines the procedures for ratio verification.

3.10 Automatic shutdown. The unit shall be equipped with a ratio metering system that is calibrated to a minimum of ± 3%. The ratio metering system shall be adjustable and contain an automatic safety cutout feature for both the "A" and "B" components.

3.11 Emergency shutoff. The unit shall contain a conspicuously mounted emergency or panic shutoff switch, which is activated by pushing with the palm of the hand. The switch shall be reset using a reset key.

3.12 Overpressure safety shutoff. The unit shall be equipped with an overpressure cutoff of either a blow plug (e.g., safety or rupture disc) or spring-loaded regulator (e.g., safety or relief valve). This system shall also be tied into the emergency shutoff switch. This shall prevent the machine from being reactivated without inspection.

3.13 Cycle counter. The unit shall be equipped with a cycle counter to record the total number of operational cycles. The counter shall not have a reset feature. If a reset feature is desired, a separate counter shall be required.

3.14 Air control valve reset. For pneumatically driven units, a means shall be provided to mechanically or pneumatically reset (shift) the metering pump air motor control valve (usually a spool valve).

3.15 Pressure. Units shall be capable of handling high viscosity, solvent free coatings with pot lives as short as 5 minutes at a pressure of not less than 5,000 psi (34.5 MPa) measured at the outlet of the metering pump. Units shall be able to meet these requirements while operating at an ambient temperature of 40 °F (4.4 °C) and providing material at the gun at a minimum temperature of 77 °F (25 °C). All units shall be capable of operating a minimum of two spray guns.

3.16 Piping. All piping, hoses, fittings, valves, and other pressure containing components shall be rated for service at or above the maximum design pressure of the system in which they are used. In other words, components exposed to pressure from the feed pumps shall be rated for service at or above the maximum design pressure of the feed pumps, and components exposed to pressure from the metering pumps shall be rated for service at or above the maximum design pressure of the metering pumps. Parts that are vented at all times (cannot be sealed to form a pressurized volume) do not have to meet this requirement.

3.17 Volume. Units shall be capable of delivering an adequate paint volume to achieve a suitable spray pattern without the use of solvents or other added viscosity reducing materials, under the following conditions:

3.17.1 The maximum number of spray guns that can be attached to the unit operating simultaneously at full capacity.

3.17.2 At an ambient temperature of 40 °F (4.4 °C), material at the spray guns shall be at a minimum temperature of 77 °F (25 °C).

3.17.3 The following configurations are to ensure that the equipment provided meets a minimum operating capability, however, with respect to current safety and engineering ratings, the manufacturer shall determine the diameter and other specifications of the hoses from the pump to the remote mixing station.

3.17.3.1 With material mixed at a remote mixing station: Unmixed material supply lines (½ inch 12.7 mm) for base component, ? inch (9.5 mm) for hardener component) of 100 feet (30.5 m) running from the metering pump to the remote mixing station and ? -inch (9.5 mm) 100-foot (30.5 m) lines (plus standard whip lines) running from the remote mixing station to each spray gun.

3.17.3.2 If equipped for mixing material at the pump (i.e., without the remote mixing station): ? -inch (9.5 mm) lines of 175 feet (53.3 m) (plus standard whip lines) running from metering pump to each spray gun.

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3.18 Whip lines. Whip lines to the guns shall be of the standard configuration to allow minimal fatigue of painters when spraying in confined spaces.

3.19 Maintenance. Units shall be capable of delivering a minimum of 1500 gallons (5.7 kL) of mixed material before any significant maintenance is required. This requirement excludes maintenance considered to be minor, such as adjustment of packing or addition of oil to throat seal cups, etc.

3.20 Physical layout. Although the physical layout of units varies between manufacturers, the following design considerations shall be followed. The unit, to include the drum handling apparatus, if ordered, shall be skid mounted or containerized. Skid mounted equipment can contain provisions for wheels for ease of movement where desired. Units can consist of either one or multi-piece systems where the optional drum feed system is either attached or detached from the main metering pump assembly. The detachable drum feed system shall be self-supporting with respect to its pumps, regulation, and agitators. Each piece of a unit shall have a lifting frame capable of supporting the entire load of the piece when fully loaded with material. The lifting frame of each piece shall be certified to comply with the requirements of NAVSEA 0989-LP-030-7000 - Lifting Standard. The entire system shall be explosion-proof. For containerized units, all equipment within the physical boundaries of the container shall be explosion-proof, including equipment auxiliary to the pumping equipment, such as lights, heaters, and ventilation fans. All skids and containerized units shall be fabricated with provisions that allow movement by forklifts, such as forklift pockets.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

5.2.1 The manufacturer of the item must have been producing a product meeting the requirements of this CID for at least 2 years.

5.2.2 The company must have sold at least 1000 units meeting this CID in the commercial market place over the past 2 years.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

AA59780	-	**XX**	**X**
CID number		Size (see code below)	Type (see code below)

Size Code		Type Code	
Size	Code	Type	Code
20-gallon capacity	20	1	1
55-gallon capacity	55	2	2

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7.2 Source of documents.

7.2.1 FAR. The Federal Acquisition Regulation (FAR) may be obtained from the Regulatory Secretariat, 1800 F Street NW, Washington, DC 20405, or online at http://www.acqnet.gov.

7.2.2 NAVSEA publications. NAVSEA publications may be obtained the Naval Logistics Library, 5450 Carlisle Pike, Mechanicsburg, PA 17055 or online at http://nll.ahf.nmci.navy.mil.

7.3 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.
b. Supporting requirements including heated or insulated lines, spray guns, repair tools, repair supplies, service manual, technical support, on-site training, training video, and a warranty package for the unit and its associated accessories.
c. Product conformance provisions (e.g., OSHA compliance).
d. Quantity required.
e. Packaging requirements.

7.4 Key words.

Agitator
Hose
Pressure-feed
Pump
Regulator
Siphon-feed
Spray Gun
Whip Line

MILITARY INTERESTS

Custodians:
Army -AR
Navy - SH
Air Force - 99

Review Activities:
Navy - AS, CG, NP, YD
Air Force - 03, 11
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Navy - SH
(Project 4940-0783-000)

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